

INTER-PROCESS COMMUNICATION USING PIPE

K K Mithul Kiruthik

241801157

Write a C program to demonstrate Inter-Process Communication (IPC) using a pipe. The program should create a parent and child process using `fork()`. The parent process should send a message to the child process through the pipe, and the child process should read and display the received message.

Input Format:

The input consists of a single integer `n` entered by the user.

Output Format:

- Displays the value sent by the parent process
- Displays the processed value (double of the number) received and computed by the child process

Sample Input:

5

Sample Output:

Parent sent: 5

Child received: 5

Child processed value (double): 10

CODE:

```
#include <stdio.h>
```

```
#include <unistd.h>
```

```
#include <string.h>
```

```

int main() {
    int fd[2];
    pid_t pid;

    char write_msg[] = "Hello from Parent Process";    char
    read_msg[100];

    // Create pipe    if (pipe(fd)
    == -1) {        printf("Pipe
    failed\n");        return 1;
    }

    // Create child process    pid
    = fork();

    if (pid < 0) {        printf("Fork
    failed\n");        return 1;
    }

    // Parent Process

    if (pid > 0) {        close(fd[0]); //
    Close read end

```

```
    write(fd[1], write_msg, strlen(write_msg) + 1);  
printf("Parent sent: %s\n", write_msg);
```

```
    close(fd[1]); // Close write end  
}
```

```
// Child Process
```

```
else {    close(fd[1]); // Close  
write end
```

```
    read(fd[0], read_msg, sizeof(read_msg));  
printf("Child received: %s\n", read_msg);
```

```
    close(fd[0]); // Close read end  
}
```

```
return 0;  
}
```

Output:

```
Enter a number: 5 Parent  
sent: 5
```

Child received: 5

Child processed value (double): 10